TANAY RAGHUNANDAN SRINIVASA

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Education

MSc in Robotics - Technische Universiteit Delft 2028
B.Tech in Robotics & Autonomous Systems - Plaksha University, Mohali 2026

Work Experience

Project Trainee, UR Rao Satellite Center, Indian Space Research Organisation Jan 2026 - May 2026
- Developed Physics Informed Neural Networks to solve the **free-terminal time** optimal spacecraft attitude control problem.
- Solved the **Hamilton-Jacobi-Bellman** Partial Differential Equation, a necessary and sufficient condition for optimality, for a time-energy objective.
- Demonstrated **robust convergence** of the proposed controller architecture through Monte-Carlo simulations from various initial conditions.

Research Intern, UR Rao Satellite Center, Indian Space Research Organisation May 2025 - Jul 2025
- Developed Physics Informed Neural Networks to solve **Infinite Horizon Optimal Control Problems** for aerospace applications.
- Applied the **Extreme Theory of Functional Connections** to solve the Inverted Pendulum and Spacecraft Detumbling Optimal Control Problems.
- Derived the Hamilton-Jacobi-Bellman Equation, a necessary and sufficient condition for optimality, for a **constrained action space** and LQR cost.

Student Tutor, Plaksha University Feb 2025 - May 2025
- Conducted tutorials and computational labs for RO2001: System Dynamics and Control, covering **system modeling** and **performance analysis**.
- Designed and assisted students with the course balancing robot project, focusing on **motor characterization**, **sensor filtering**, and PID control.

Summer Research Intern, Robert Bosch Center for Cyber-Physical Systems, IISc May 2024 - Aug 2024
- Designed and Tuned a Linear Controller with a **Gain Scheduler** to Balance **BiSteering Two Wheeled Robot**, achieving balance for 76s.
- Researched and Measured Loaded and Unloaded **Motor Deadband** and Compensation techniques.
- Measured and Compared Settling Time and Overshoot of the **Bosch Sensor Fusion Algorithm** and a **Kalman Filter**.

Projects

Degradation Aware Optimal Battery Cycling for Smart Grids | *Prof. Vivek Deulkar* Aug 2025 - Jan 2026
- Developed RL agents to optimally cycle heterogeneous battery banks to fulfill demand-supply mismatches in **renewable energy generation**.
- Designed an **off-policy function-approximation** kernel inspired by the Extreme Learning Machine, outperforming handcrafted features by 2x.
- Researched and Integrated rainflow-based lithium ion depth-of-discharge **battery degradation** models to minimize operational costs.

SegwayRL | *Prof. Sandeep Manjanna* Jan 2025 - May 2025
- Designed and trained **PPO** and **DQN** agents to balance a segway robot, with training in both simulation and hardware.
- Developed a segway training and deployment environment using **OpenAI Gymnasium**, designed for 10ms sensing-actuation loop in hardware.
- Designed a **state-action reward function**, to penalize states far away from equilibrium, while penalizing large control efforts for small errors.

Low Altitude Remote Sensing (LARS) UAV for Crop Health Monitoring | *Prof. Sunita Chauhan* Jan 2024 - May 2025
- Developed a **ROS based communication** protocol to interface UAVs with ground robots to enable real-time autonomous crop monitoring.
- Developed a lightweight convolutional neural network based on MobileNet-v4 meant to be deployed on **constrained edge devices**.
- Calculated **Ground Sampling Distance (GSD)** at varying altitudes for UAVs, benchmarking imaging capabilities for precise crop health monitoring.

Publications

Nanda, T.R., Shukla, A., Srinivasa, T.R., Bhargava, J., Chauhan, S. (2025). *Advancing Real-Time Crop Disease Detection on Edge Computing Devices Using Lightweight Convolutional Neural Networks*. In: Arai, K. (eds) Intelligent Systems and Applications. IntelliSys 2025. Lecture Notes in Networks and Systems, vol 1567. Springer, Cham. https://doi.org/10.1007/978-3-032-00071-2_33 .

T. R. Srinivasa and S. Kumar, *Solving Infinite-Horizon Optimal Control Problems using the Extreme Theory of Functional Connections*, 2025 Eleventh Indian Control Conference (ICC), Bangalore, India, 2025, pp. 147-152. https://doi.org/10.1109/ICC69100.2025.11372426

S. Kumar and T. R. Srinivasa, *HJB Optimal Feedback Guidance for Spacecraft Attitude Manuever via Physics-Informed Neural Networks*, AIAA SciTech Form 2027 (Accepted Aug 31, 2026).

Skills

Programming Languages: Python, MATLAB, C/C++, ROS2, Bash, LaTeX.
Design and Manufacturing: Fusion 360, RD Works.
Micro-controllers: Raspberry Pi 3B+/4B/5, ESP-32, Arduino Uno/Nano, Teensy 4.1, STM32F103RB.

Positions of Responsibility

Mechanical Workshop Coorndinator, Robotics Lab, Plaksha University Aug 2024 - Jan 2025
Developed the floor plan, designed the workbench, procured tools, and set-up the mechanical workshop.

Achievements

Academic Achievement Award Jul 2026
Awarded for achieving the highest CGPA in the Robotics & Autonomous Systems Major in the graduating B.Tech Class of 2026.

Dean's List Nov 2025
Awarded for outstanding academic performance during the Academic Year 2024-2025.